Assignment Set 2

CS-423 (Artificial Intelligence)

PART -IV (Total Marks 3*10=30)

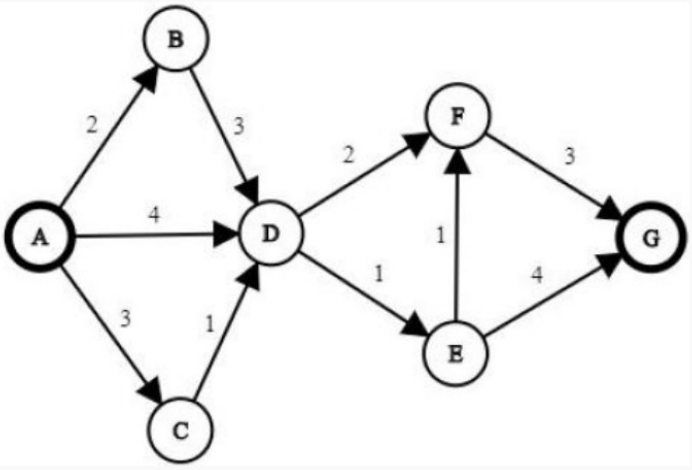
1) If *h1* and *h2* are two admissible heuristics, then which of the following are guaranteed to be admissible?

- [x] min(h1,h2)
- [x] max(h1,h2)
- [x] (h1+h2)/2
- [x] (h1.h2)^0.5

2) If *h1* and *h2* are two admissible heuristics, then which of the following will give a heuristic that dominates both *h1* and *h2*?

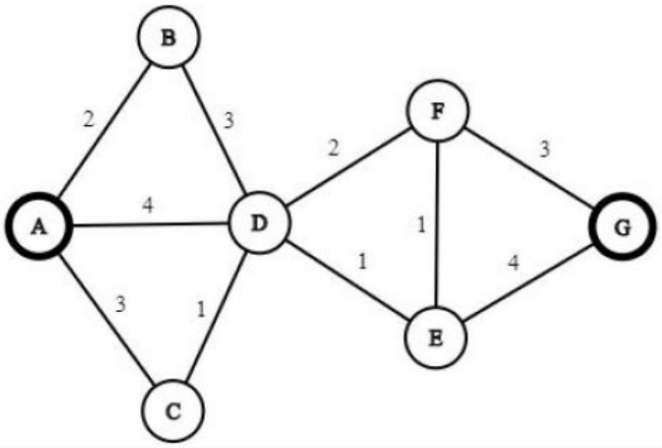
- [ ] min(h1,h2)
- [x] max(h1,h2)
- [ ] (h1+h2)/2
- [ ] (h1.h2)^0.5

3) Consider the following directed graph in which we are searching from start state A to goal state G. The number over each edge is the transition cost. Additionally, we are given a heuristic function h as follows {h(A)=7, h(B)=5, h(C)=6, h(D)=4, h(E)=3, h(F)=3, h(G)=0}. The given heuristic function is:



- [ ] Admissible as well as consistent
- [x] Admissible but not consistent
- [ ] Consistent but not admissible
- [ ] Neither admissible nor consistent

4) Consider the following undirected graph in which we are searching from start state A to goal state G. The number over each edge is the transition cost. Additionally, we are given a heuristic function h as follows {h(A) = 7, h(B) = 5, h(C) = 6, h(D) = 4, h(E) = 3, h(F) = 3, h(G) = 0}. Find the sequence of nodes explored (i.e., the sequence of removals from its frontier) by A* graph search algorithm. Assume that, in case of ties, the search procedure uses a lexicographical order for tie-breaking. Write the answer as a string with no spaces. For example, if the order of exploration is A followed by B followed by C followed by D then write ABCD.



ABDECFG

5) Which of the following algorithms is/are guaranteed to give an optimal solution?

- [ ] Greedy Best First Search
- [x] A* with zero heuristic
- [x] A* with consistent heuristic
- [ ] Depth First Search

6) Consider an infinite search space $\mathbb{Z} \times \mathbb{Z}$. The start state is at (0, 0) and the goal state is at (g_x, g_y). Given that the agent can move from a state (x, y) to either of {(x+1, y), (x-1, y), (x, y+1), (x, y-1)} with a unit step cost, find the number of nodes explored using A* tree search with Manhattan distance heuristic for $(g_x, g_y) = (6, 7)$. Assume no duplicate detection.

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7) Which of the following statement(s) is/are correct?

- [x] A* graph search expands no more nodes than uniform-cost graph search.
- [ ] Uniform cost search is a special case of best-first tree search.
- [ ] A* graph search expands no more nodes than depth-first graph search.
- [ ] Weighted A* implementation does not need a priority queue.

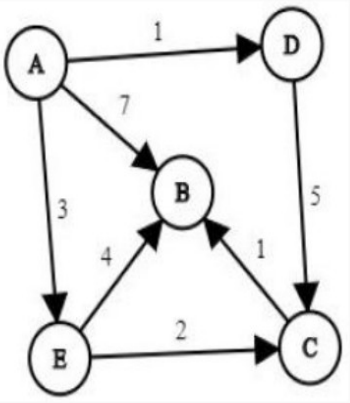
8) Consider the A* tree search algorithm with $f(n) = (2 + x).g(n) + (x^2).h(n)$ where g(n) is the cost from the start state to state n and h(n) is an admissible estimate of the cost from state n to the goal state. For how many integer values of x is the algorithm guaranteed to be optimal?

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9) Consider a heuristic function $h_2(n) = 3.h_1(n)$ where $h_1(n)$ is admissible. Which of the following statement(s) is/are correct?

- [ ] $h_2(n)$ is admissible.
- [ ] A* graph search with h_2 will return the optimal path.
- [x] A* tree search with h_2 will return a path at most thrice as long as the optimal path.
- [ ] $h_2(n) + 1$ is guaranteed to be inadmissible for any h_1.

10) Consider the following directed graph in which we are searching from start state A to goal state B. The number over each edge is the transition cost. Additionally, we are given a heuristic function h as follows {h(A) = 5, h(B) = 0, h(C) = 1, h(D) = 6, h(E) = 2}. Find the sequence of nodes explored by Depth-First Branch and Bound Search Algorithm. A node is explored further only if $g(n_k) + h(n_k) <$ bound. Assume that the search procedure uses a lexicographical order for exploration/tie-breaking. Write the answer as a string with no spaces. For example, if the order of exploration is A followed by B followed by C followed by D then write ABCD.



ABDEBCB